



CLOSED-LOOP BIO-SOLAR BLUEPRINT

Micro-Grid Food & Energy Independency

Presented by: **Sipho Wiseman Khuzwayo**, Founder & Managing Director



The "Triple Threat" Risk Gaps

Traditional agricultural business models inside Mpumalanga suffer from three distinct operational risks that bankrupt facilities and scare off smart capital: **High Mortality Rate** | **High Operational Cost** | **High Selling Price**.



Utility Tariffs

Skyrocketing seasonal utility power costs drain up to 35% of standard farm bottom-lines.



Highveld Frosts

Winter cold snaps trigger massive live inventory hypothermia, raising mortality to 18%.



Open-Herd Viruses

Standard open single-room designs allow pathogens to wipe out entire livestock assets.



The Closed Loop Bio-Solar Engine

Our solution integrates high-yield biology with localized co-generation. By using every byproduct on-site, we decouple operations from vulnerable public grids: **CO₂ capture** | **Waste recycling** | **Water retention**.

- ✓ **30kW BESS & Solar Rooftops:** Powers daytime boreholes, automated augers, and primary controls directly.
- ✓ **Methane Co-Gen Engine** Recycle swine/poultry slurry into biogas, co-generating supplementary night-time volts & heat energy.
- ✓ **Sub-Floor Thermal Plumb:** Routes engine exhaust heat directly underneath nursery pods to warm animals.



Zero-Mortality-Microclimate-Enclosures

Passive Convection

Designed with an 18-degree roof pitch to naturally siphon warm, stale air out, bypassing heavy electric fans entirely.



Mpumalanga Industrial Food Gaps

**ZAR
1.2B**

**TAM (Total Addressable
Market)**

Bottoms-Up Sizing Blueprint

The Secunda-Kinross development corridors hold dense worker populations with high demand for premium retail meat supply:

- ✓ **Total Addressable (TAM):** ZAR 1.2 Billion regional mining meat consumption profile.
- ✓ **Serviceable Addressable (SAM):** ZAR 180 Million high-grade wholesale off-take demand.
- ✓ **Serviceable Obtainable (SOM):** ZAR 3.39 Million Initial baseline year capture.



Double High-Velocity Income Model

Biological Revenue Pillar	Standard Batch Sizing	Target Wholesale Price	Annual Revenue Run (Baseline)
Poultry Sprint Barn	5,760 Broilers over 6 waves/yr = 34 560/yr	ZAR 38.00 / kg (at 1.77kg harvest)	ZAR 2,324,500
Swine Grower Pipeline	160 Pigs over 3 waves/yr = 480/yr	ZAR 32.00 / kg (at 70kg carcass)	ZAR 1,075,200
Organic Bio-Solids	Continuous digester byproduct	ZAR 65.00 / 10kg processed bag	Auxiliary Cash Engine



Supply & Off-Take Strategy

Lwandile Engineering Projects bypasses volatile middle-man logistics by locking down direct B2B institutional relationships:



Direct Off-Take MoUs

Drafting wholesale purchase agreements with regional meat processing networks inside Govan Mbeki Municipality.



Supply Chain

Securing strategic food-service provider contracts for large worker hubs in Secunda & surroundings



Precision Alliance

Partnering with regional NPOs and Mam Dube coop retail points for immediate municipal retail distribution.



Competitive Insulated Edge



Traditional Grid Farms

- ✓ Heavy exposure to rising Utility tariffs.
- ✓ No backup thermal heating, leading to high swine & poultry mortality.
- ✓ Open housing leads to single-outbreak biological wipes.



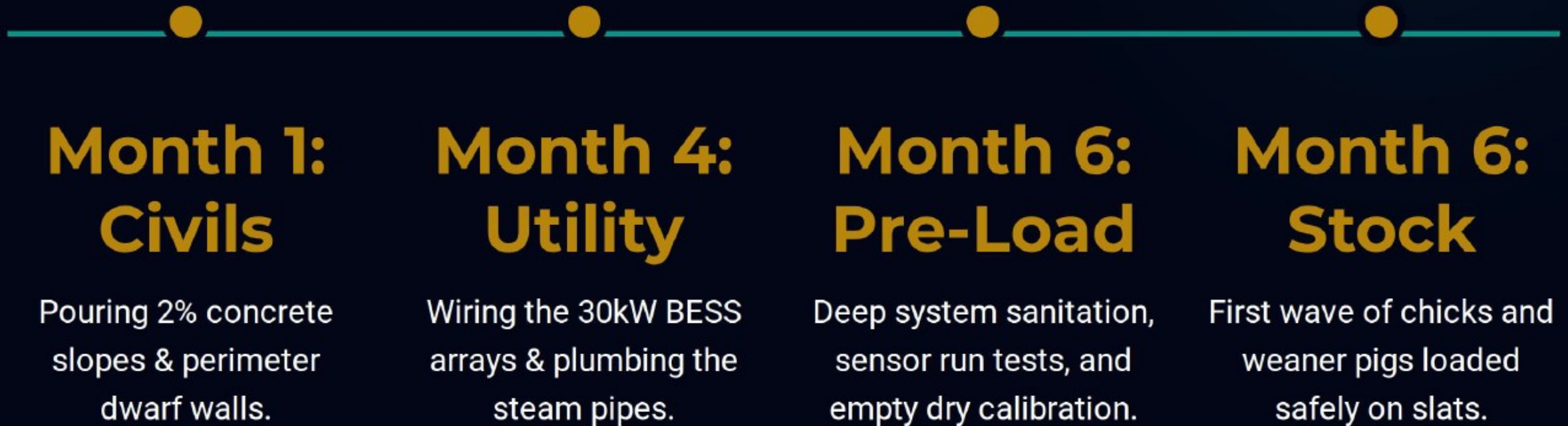
Lwandile Shield System

- ✓ Zero-utility cost shield locked at ZAR 1.80/kWh.
- ✓ Recycled biogas engine steam provides natural sub-floor warmth.
- ✓ Physical 4-pod air-gaps limit viral risk down to a 25% boundary, in pursuit of zero-mortality.



Development Roadmap & Gates

The seed





3-Year High-Velocity Cash Forecast



Target EBITDA

Annual operating cash gains show robust stability, completely insulated by our Zero-Grid utility cost matrix:

- ✓ **Year 1 EBITDA:** ZAR 1,180,000
- ✓ **Year 2 EBITDA:** ZAR 1,287,000
- ✓ **Year 3 EBITDA:** ZAR 1,406,600



Precision Execution Team



Cross-Boarder Strategic Alliance



PROPRIETARY INTELLECTUAL PROPERTY - Lwandile Engineering

PRESTIGE COSMIC VENTURE

15+ Years Technical Leadership

Sipho leads Lwandile Engineering Projects with extensive operational expertise across precision engineering, high-voltage microgrid architectures, and automated industrial mechanics through long-life research.

- ✓ **Ecosystem Pioneer:** Designed the closed-loop bio-thermal energy flow blueprint.
- ✓ **Socioeconomic Alignment:** Deep regional execution links with municipal cooperatives in Kinross and Secunda.

CONNY



Capital Requirement & Payback

Total Baseline Request: ZAR 3.1M

Every single Rand maps directly to a high-value physical asset engineered to shield and scale biological production metrics:

- ✓ **Civils & Structural Enclosures:** ZAR 1.22M (insulated barns and slatted floors).
- ✓ **30kW/30kWh Solar BESS Core:** ZAR 1.07M (Hybrid inverters, batteries, PV modules, 70 cubic meter **anaerobic bio-digester**)
- ✓ **Hydrology & Biosecurity:** ZAR 0.81M (boreholes, UV filters, shower locks, IoT & gravity water tank.



2.6 Years

Investment Payback Period

Working Capital & Reserve: Embedded as 10% contingency of the total capital.



ASITHOKOZE- LET US EXECUTE

Sipho Wiseman Khuzwayo | Founder & MD

siphokhuzwayo12@gmail.com

<https://www.linkedin.com/in/sipho-khuzwayo>

🗨️ CLOSING SUMMARY

"We are offering a secure, structured, and bankable opportunity to back a resilient green food infrastructure. The engineering is finalized, the biosecurity is locked, and the regional partners are ready. Join us in building Mpumalanga's premier off-grid sovereign food & bio-energy closed-loop agricultural network."